

GIHEN DHARMARATHNE

Data Science Undergraduate

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PROFESSIONAL SUMMARY

Highly motivated Data Science undergraduate at General Sir John Kotelawala Defence University with hands-on experience in machine learning, computer vision, and NLP. Proven ability to build end-to-end analytical solutions, from real-time sports analytics dashboards and predictive time-series models to custom YOLO-based object detection applications. Passionate about transforming complex datasets into actionable insights and deploying user-centric web applications..

TECHNICAL SKILLS

Programming Languages: Python, SQL, HTML, CSS

Machine Learning & AI: Computer Vision (YOLO), Natural Language Processing (NLP), Time Series Analysis, Predictive Modeling

Libraries & Frameworks: Streamlit, Pandas, NumPy, Scikit-Learn, OpenCV

Data & Analytics: Web Scraping, Exploratory Data Analysis (EDA), Sports Analytics, Interactive Dashboards

Tools & Deployment: Git, GitHub, GitHub Pages

PROJECTS

Automated Wide Ball Detection System

Python, OpenCV, YOLO, Computer Vision

[GitHub Repo](#)

Engineered a computer vision system utilizing object detection algorithms to automate the identification of wide balls in cricket match footage.

Processed video data to track ball trajectory relative to the crease, aiming to reduce human umpiring error and enhance decision-making accuracy in real-time.

Heart Disease Prediction Model

Python, Scikit-Learn, Pandas, Machine Learning

[GitHub Repo](#)

Developed a predictive classification model to assess heart disease risk based on clinical parameters, optimizing for high sensitivity and accuracy.

Conducted extensive exploratory data analysis (EDA) and feature engineering to identify key health indicators contributing to cardiovascular conditions.

CricMiner: Cricket Analytics Application

Python, Web Scraping, Streamlit, Pandas

[GitHub Repo](#)

Architected a full-stack data mining application to scrape and synthesize player performance statistics from major cricket databases.

Implemented interactive visualization modules to analyze player career trajectories, assisting in data-driven team selection and talent scouting.

Matchup Simulator

Statistical Modeling, Data Simulation

[GitHub Repo](#)

Built a probabilistic simulation engine to forecast batter-vs-bowler outcomes specifically for Sri Lankan cricket players using historical matchup data.

Modeled various in-game scenarios to provide strategic insights for coaching staff regarding bowler rotation and field placement optimization.

LPL Death Overs Batting Analysis

Data Visualization, Strategic Analysis, Power BI, Python

[GitHub Repo](#)

Analyzed Lanka Premier League (LPL) data to evaluate batting efficiency during death overs, identifying high-impact players and scoring trends.

Produced actionable reports on strike rates and boundary frequencies to bridge the gap between current batting strategies and optimal run-scoring targets.

2026-T20-World-Cup-Analysis

Data Visualization, Strategic Analysis, Power BI, Python

[GitHub Repo](#)

Designed a dynamic dashboard to track team and player statistics across the T20 World Cup, highlighting key performance indicators (KPIs).

Synthesized complex tournament data into intuitive visual formats, enabling users to compare team strengths and weaknesses instantly.

Policy Analysis & Adaption System

Python, NLP, Pandas, Large Language Models (LLMs)

[GitHub Repo](#)

Developed an automated system utilizing Natural Language Processing (NLP) to analyze, extract, and summarize key insights from complex policy documents.

Implemented adaptive algorithms to dynamically track policy changes and generate data-driven compliance recommendations for stakeholders.

Live Cricket Score Dashboard

Python, API Integration, Streamlit

[GitHub Repo](#)

Built an interactive dashboard (Cricket Score Dashboard) that fetches and visualizes match data, focusing on run rates, wagon wheels, and partnership statistics.

Implemented real-time data handling to provide instant updates and comparative visualizations for ongoing matches, enhancing the viewer experience.

EPC Energy Analysis Dashboard

Power BI, DAX, Data Modeling, Data Visualization

[GitHub Repo](#)

Developed an interactive Power BI dashboard to analyze Energy Performance Certificate (EPC) data, tracking building energy efficiency and carbon emission metrics.

Implemented complex DAX measures and optimized data models to process large-scale energy datasets, providing actionable visual insights for energy optimization.

Road Safety Analysis Dashboard

Power BI, SQL, Data Visualization, Data Cleaning

[GitHub Repo](#)

Developed an interactive dashboard to analyze road accident incidents, identifying high-risk locations, casualty trends, and contributing factors such as weather and road conditions.

Transformed raw accident data into a series of visual reports, utilizing geospatial mapping and time-series analysis to provide data-driven insights for improving road safety measures.

CERTIFICATIONS

Rajasthan Royals School of Business: Sports Analytics and Management Certification

EDUCATION

General Sir John Kotelawala Defence University
B.Sc. in Applied Data Science Communication

Ratmalana, Sri Lanka

Bandaranayake College

Gampaha, Sri Lanka

LEADERSHIP & ACTIVITIES

Social Media Analyst – DatAInspire.KDU

Jan 2025 – 2026

General Sir John Kotelawala Defence University

Ratmalana, Sri Lanka

Directing digital engagement strategies for the Club, including content scheduling and real-time audience trend tracking for events and campaigns.

Synthesized monthly performance reports for the executive committee, ensuring data-driven decision-making in content timing and platform selection.

Collaborated with stakeholders to align promotional activities with organizational goals, maintaining consistent brand visibility to drive future audience growth.

Member | University Cricket Team

2024 – 2026

General Sir John Kotelawala Defence University

Ratmalana, Sri Lanka

Representing the university in inter-university tournaments, demonstrating teamwork, discipline, and high-pressure decision-making skills.

Actively participating in rigorous training sessions and tactical planning to enhance team performance and maintain peak physical fitness.

Assistant Coach | MB Cricket Academy

2023 – Present

Negombo, Sri Lanka

Mentoring young athletes on cricket fundamentals, focusing on technique development, game awareness, and sportsmanship.

Assisting in the design and execution of specialized training drills for batting and bowling to improve individual and team performance.

Monitoring player progress and providing constructive feedback to help students achieve their athletic goals.